

Rohan Kale

United States Citizen | 408-431-8043 | kalerohan42@gmail.com | linkedin.com/in/rohankale42 | github.com/Rohan-Kale | rohan-kale.github.io

EDUCATION

The Pennsylvania State University

Aug 2024 – May 2028

B.S. in Computer Science

GPA: 3.55/4.0

Awards/Honors: Dean's List (Top 5% of class)

Relevant Coursework: Computer Architecture, Operating Systems, System Programming, Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Probability & Statistics

EXPERIENCE

CrowdStrike | Software Engineer Intern

May 2026 – Aug 2026

- Designed and implemented a Kafka-based event signaling system in Go, enabling real-time audit event emission across 2,400+ existing customer workflows.
- Extended event emission to all firewall mutation paths across 13 REST endpoints and 4 domain entities, generating approximately 18,000 protobuf events per day across 5 cloud regions.
- Diagnosed and resolved a Go data race causing nondeterministic test failures and inconsistent event ordering, enforcing deterministic emission across 15 event sites and restoring reliable test execution.
- Eliminated nil-dereference panics in customer-facing policy CRUD paths by enforcing non-nil object invariants, removing 27 redundant defensive checks across the codebase.

Jetson.app | Software Engineer Intern

June 2024 – August 2024

- Built a React and Node.js quiz system backed by REST APIs that sampled non-repeating questions from a database-backed question bank and persisted user responses and scores for a financial literacy platform serving 10,000+ users.
- Accelerated Docker builds by restructuring image layers and caching dependencies across CI runs, reducing pipeline build time from 12 to 7 minutes.
- Streamlined randomized question retrieval by indexing frequently queried question-bank fields and consolidating database lookups from 4 queries to 1 per quiz, reducing redundant database access.

PROJECTS

PrivateAI Gateway | [GitHub] | Go, Python, Docker

September 2026

- Engineered a Go LLM gateway enforcing configurable allow, redact, tokenize, and block policies for PII and credentials, with encrypted, reversible tokenization to restore authorized values in responses.
- Implemented Redis-backed asynchronous processing with three concurrent workers, completing 142,813 requests without errors during a 15-minute, 200-client load test against a simulated model provider.
- Instrumented service health and queue latency with Prometheus; verified alert delivery within 20 seconds across 12 controlled worker, detector, and datastore failure trials.
- Developed Python workflows for issue triage and pull request checks, Kubernetes deployment manifests, and Ansible roles for Linux worker provisioning, networking, filesystem mounts, and KVM configuration.

Kale Space Program | [GitHub] | Rust, wgpu, winit

June 2026

- Developed a Rust-based 3D planetary simulation engine using wgpu and GPU compute shaders, simulating 20,000+ celestial bodies in real time at 60 FPS through GPU-accelerated physics and instanced rendering.
- Accelerated gravitational force computation using WGSL compute shaders and Barnes-Hut spatial approximation, reducing physics-step latency from 62 ms to 14 ms at 5,000 bodies.
- Designed a GPU-resident physics pipeline using storage buffers and bind groups to update thousands of body states in parallel, reducing CPU-GPU data transfers by 80% while minimizing per-frame synchronization.
- Built Python batch processing for 10,000 sensor replay jobs to evaluate simulation outputs across 3 seeds, supporting repeatable validation of simulation workloads.

LEADERSHIP

Society of Asian Scientists and Engineers | Events Director

May 2025 – May 2026

- Led a team of 6 members to plan and execute 8 technical workshops and professional development events, serving 150+ students and increasing average attendance by 30% through targeted event planning.
- Coordinated with 5 industry professionals and 3 student organizations to deliver engineering career panels and mock interviews, connecting 60+ students with mentorship and recruiting preparation.
- Introduced a shared planning tracker to assign ownership, monitor deadlines, and coordinate event logistics across 4 committees, reducing last-minute scheduling conflicts by 25%.

SKILLS

Languages: Go, Rust, Python, JavaScript, SQL, WGSL, Java, C++

Infrastructure & Data: Kafka, PostgreSQL, Docker, Linux, SQLite, Kubernetes, AWS, GCP, Redis, Protocol Buffers, REST APIs, CI/CD

Frameworks & Tools: FastAPI, Flask, Node.js, React, wgpu, winit, Git, GitHub, BitBucket, Github Actions, Jenkins, Wireshark, hping3, Claude, Codex